College Information Report

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ABSTRACT
This project uses multiple linear regressions to predict total student enrollment at U.S. higher education institutions using the 2013 IPEDS dataset. After cleaning and preparing the data, including log-transformation the target variable to address skewness, two feature sets were tested across four experimental designs with different train test splits. Feature Set A including basic numerical predictors, while Feature Set B added institutional characteristics such as a degree level and control type. Model performance was evaluated using R^2, RMSE, and MAE. The best model achieved an R^2 of approximately .78 on test data, indicating strong predictive performance. Results show that institutional characteristics, especially the highest degree offered and public vs. private status are key drivers of enrollment. The findings demonstrate that a relatively simple regression model can effectively explain variation in enrollment using publicly available data.

I. INTRODUCTION

This project analyzes institutional data from the Integrated Postsecondary Education Data System to predict total student enrollment at U.S. colleges and universities. The dataset contains over 1500 institutions and included variables related to admissions, tuition, student demographics, and institutional characteristics.

The primary objective is to develop a regression model that predicts total enrollment using a set of explanatory variables. Because enrollment varies widely across institutions, ranging from 100 to 70000+ students, a log transformation is applied to stabilize the distribution. This project evaluates multiple models using different feature sets and training splits to determine which combination produces the most accurate predictions.

II. BACKGROUND

IPEDS is a comprehensive data collection system maintained by the National Center for Education Statistics. It gathers annual data from postsecondary institutions in the United States, including information on enrollment, admissions, finances, and student outcomes. The dataset is widely used by policymakers, researchers, and institutions to analyze trends in higher education.

Understanding what drives institutional enrollment is important for several reasons. Enrollment size affects funding, staffing, campus infrastructure, and strategic planning. Larger institutions tend to have more resources and broader program offerings, while smaller institutions may focus on specialized education. By identifying the factors associated with enrollment size, this analysis helps provide insight into structural differences across institutions and supports data-driven decision making in higher education.

III. EXPLORATORY ANALYSIS

This section will be similar to your exploratory analysis project. First, provide a summary of the data set similar to your first exploratory analysis: *e.g. this data set contains 398 samples with 7 columns with various data types*. In this summary, provide the data types of your columns (in a table) and then rather than providing tabular statistics and plots for each variable, provide only statistics and plots that seem unusual. For example, if one or two variables have significant missing values or the distribution of the variable is skewed or looks unusual note that. Provide the unusual statistics or plots in this section. Provide any other appropriate plots (e.g. correlation matrix, heatmaps, bar charts, etc.) that you deem necessary.

The dataset contains 1534 samples with 145 variables, including numerical and categorical variables.

Table 1: Data Types

Variable Name	*Data Type*
Applicants total	Float64

Admissions total	Float64
Percent admitted- total	Float64
Tuition and fees	Float64
SAT Math 75th percentile	Float64
Graduation rate (6 Years)	Float64
Percent receiving Pell grants	Float64
Highest degree offered	Object
Control of institution	Object
Total enrollment (Target)	Float64

Figure 1:

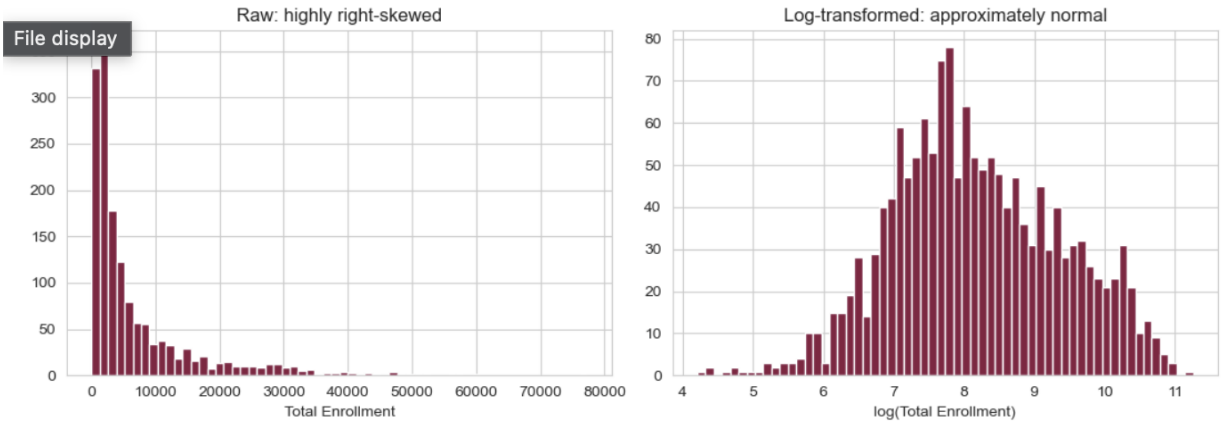


Figure 2:

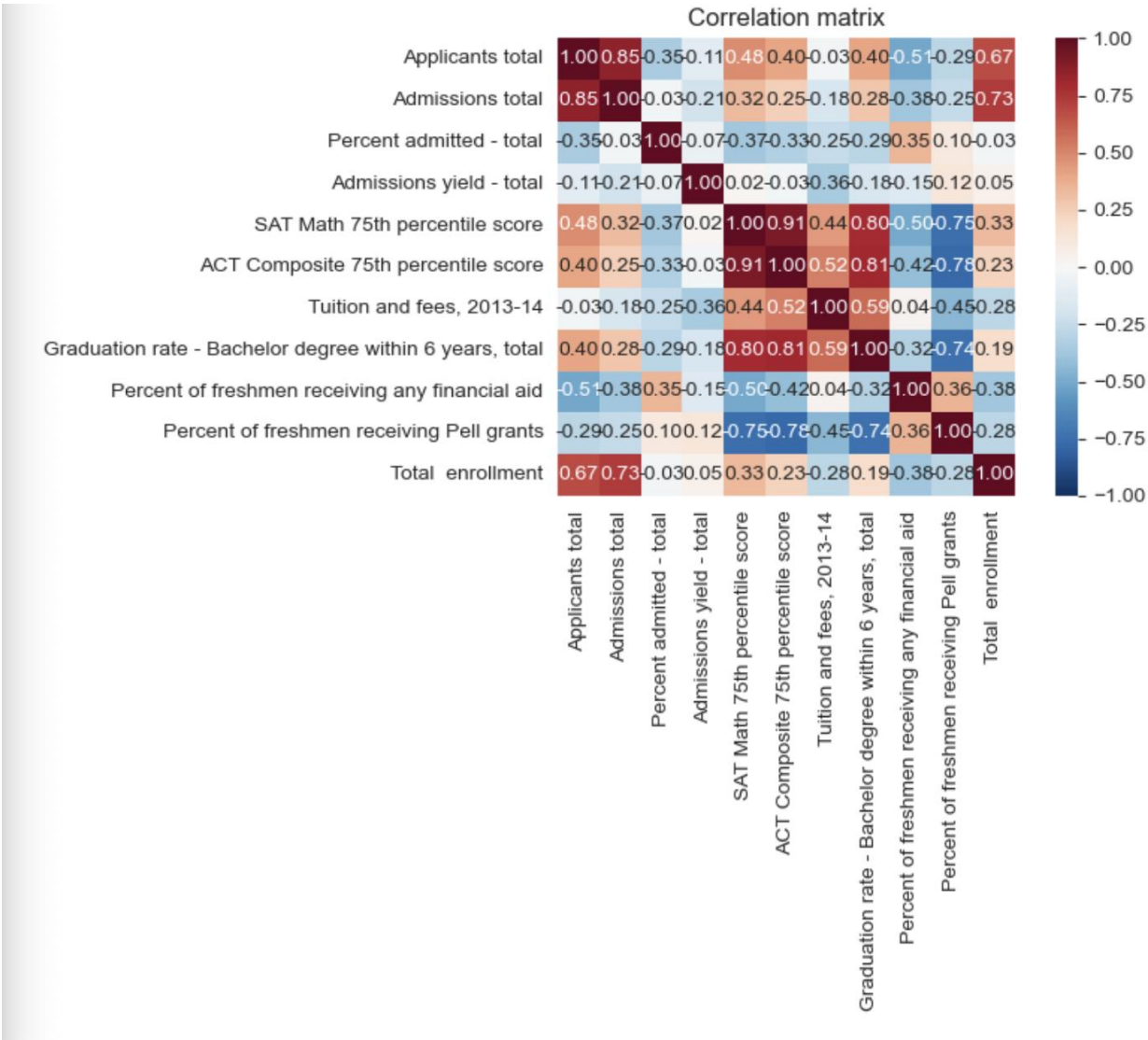


Figure 3:

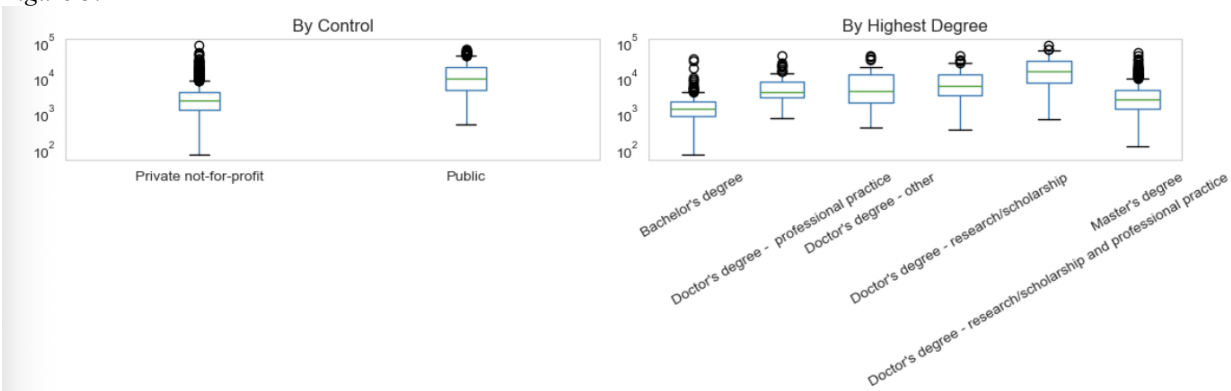


Figure 4:

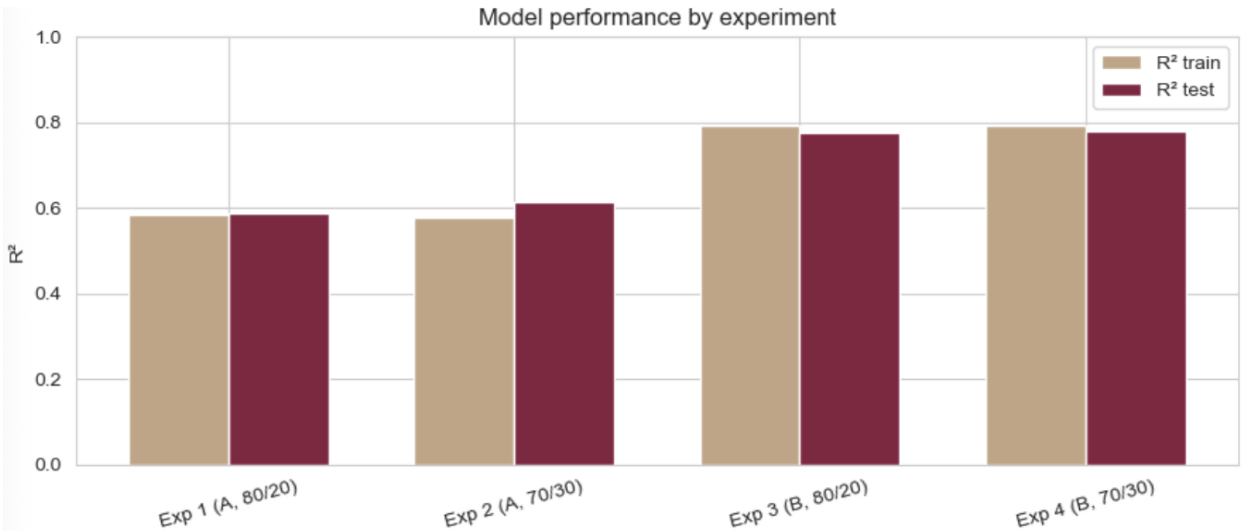
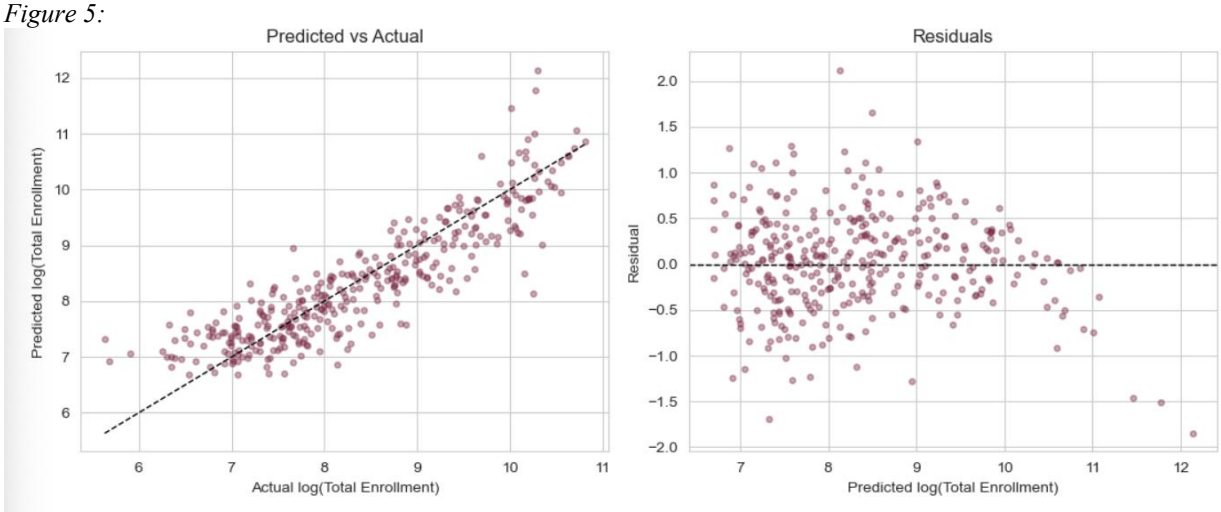


Figure 5:



IV. METHODS

In this section, describe how you prepared the data for your model and performed multiple experiments using different parameters for the model(s).

A. Data Preparation

Describe how you prepared the data for your model. For example, you might need to normalize the data, so variables with wider ranges of values don't overshadow variables with smaller ranges. If you decide to drop variables from the model or create variables from existing columns, explain the process and the reasoning behind those decisions.

We went through several steps to prepare the data including; dropping row with missing target values, applying log transformation to total enrollment, removed variables that directly measure enrollment to avoid data leakage, selected 2 feature sets, dropped rows with missing predictor values, one hot encoded categorical variables, standardized all predictors using StandardScaler.

B. Experimental Design

You will run your model several times with different parameters to see what different results you get. In a table, describe your experimental parameters. Three or four experiments are sufficient. This is where you will describe how you divided your data into train, validate and test data sets. For example:

Table X: Experiment Parameters

Experiment Number	Parameters
1	Feature set A with 80/20 split for train, and test
2	Feature set A with 70/30 split for train, and test
3	Feature set B with 80/20 split for train, and test
4	Feature set B with 70/30 split for train, and test

C. Tools Used

Describe all of the software tools you used to perform your data preparation and model implementation. For example:

The following tools were used for this analysis: Python v3.5.2 running the Anaconda 4.3.22 environment for Apple Macintosh computer was used for all analysis and implementation. In addition to base Python, the following libraries were also used: Pandas 0.18.1, Numpy 1.11.3, Matplotlib 1.5.3, Seaborn 0.7.1, SKLearn 0.18.1, and Patsy 0.41. Provide a brief explanation of why you chose these tools.

These tools were chosen because they provide efficient data handling, visualization, and machine learning capabilities widely used in data science.

V. RESULTS

A. Classification Measures/ Accuracy measure

Experiment	R^2 test	RMSE	MAE
Exp 1	.588	64469	9950
Exp 2	.613	45819	7032
Exp 3	.776	8046	3243
Exp 4	.780	11892	3678

B. Discussion of Results

Feature Set B clearly outperformed Feature A. Adding institutional characteristics such as highest degree offered and control type significantly improved model accuracy. The best model (Exp 4) achieved and R^2 of .78, indicating that it explains 78% of the variation in enrollment.

The worst preforming models were those using feature set A, which lacked key categorical variables. These models were unable to capture structural differences between institutions.

C. Problems Encountered

Several challenges arose during the project, the target variable had formatting issues, significant missing data in SAT/ACT variables reduced sample size, multicollinearity between predictors required variable selection, and finally skewed distributions required transformation.

D. Limitations of Implementation

Linear regression assumes a linear relationship, which may not fully capture real world complexity. Complete case analysis excludes many institutions. The model is based only on 2013 data and may not reflect current trends. Large institutions are slightly under predicted. A possible better model could be Random Forest Regression.

E. Improvements/Future Work

Some future improvements could include using nonlinear models, including multiple years of data, adding geographic and regional variables, testing interaction effects between variables, and applying imputation methods instead of dropping missing data.

VI. CONCLUSION

This project demonstrates that multiple linear regression can effectively predict institutional enrollment using publicly available data. By applying a log transformation and incorporating both numeric and categorical predictors, the model achieved strong performance with an R^2 of approximately .78.

The most important predictors were institutional characteristics, particularly the highest degree offered and whether the institution is public or private. While the model preforms well overall, improvements could be made by using more advanced machine learning techniques and additional data. Overall, the model provides a useful and interpretable approach to understanding the factors that drive enrollment in higher education.

REFERENCES

-National Center for Education Statistics
-Scikit Learn Documentation
-Pandas
-NumPy
-Matplotlib
-Seaborn

Other directions:

1. 10-pt, Times New Roman, 1” margins all around (if you use this template you are already set).
2. **Ensure all tables and figures are numbered appropriately and referenced in the text.** See examples above and below.

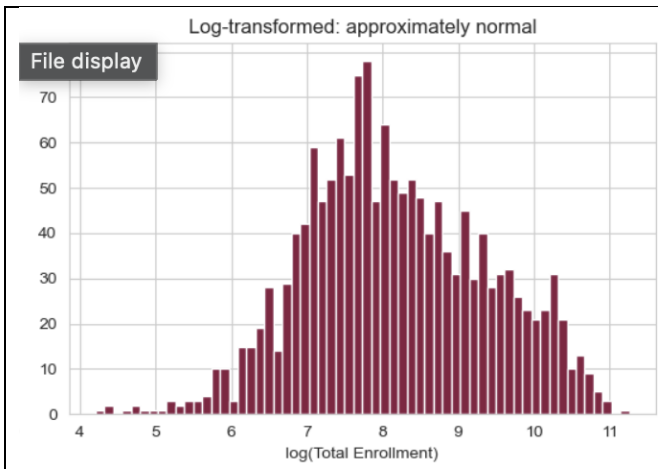


Figure 1: Comparison of X/Y from dataset (single plot) (8 pt.)

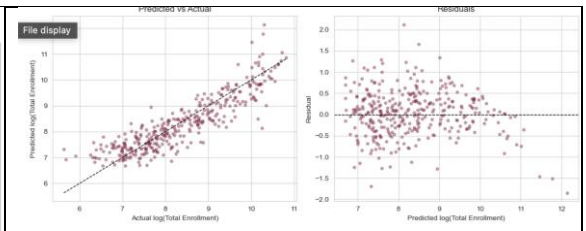


Figure 2: (a) Function Output (b) A against B (multiple plots) (8 pt.)